CASE STUDY SPECIFICATION & ANALYTICAL FRAMEWORK

Exploratory Data Analysis of the Flavors of Cacao Dataset Using R

Investigating cocoa percentages, bean origins, manufacturer dynamics, and expert quality ratings

DOMAIN	PRIMARY TOOL	FOCUS	TARGET OUTCOME
Agribusiness & Food Analytics	R & Tidyverse Environment	Exploratory Data Analysis (EDA)	Quality Sourcing Optimization

1. INTRODUCTION

The global chocolate industry represents one of the largest sectors within the food and beverage domain. Manufacturers source cocoa beans across diverse geographical regions, crafting products that vary widely in cocoa concentration, origin, bean variety, and expert quality ratings. Understanding the intricate drivers of chocolate quality offers immense strategic value to cocoa producers, artisan manufacturers, retailers, researchers, and consumers alike.

This case study details an Exploratory Data Analysis (EDA) of the **Flavors of Cacao** dataset using the **R programming environment**. By examining relationships between cocoa concentration, bean provenance, company location, bean variety, and expert ratings, the study provides evidence-based insights into what defines top-tier chocolate products.

2. BUSINESS PROBLEM

Chocolate manufacturers and procurement specialists source cocoa beans from numerous global origins. However, it remains challenging to pinpoint which structural attributes—such as cocoa percentage, geographic origin, or bean variety—contribute most significantly to high expert ratings. Business leaders require data-driven intelligence to refine sourcing strategies, optimize formulations, and enhance quality assurance frameworks.

This project addresses the challenge of identifying key characteristics associated with premium chocolate bars through systematic analytical exploration of historical expert review data.

3. AIM OF THE STUDY

The primary aim of this project is to execute an exploratory data analysis on expert chocolate review data in R to uncover patterns, correlations, and temporal trends that govern expert quality ratings.

4. PROJECT OBJECTIVES

- **Dataset Import & Setup:** Ingest the Flavors of Cacao dataset into R and establish an analytical workspace.
- **Data Cleaning & Transformation:** Clean, standardize, and preprocess variable names, data types, and textual fields.
- **Quality & Profile Assessment:** Evaluate missing values, distribution shapes, and variable profiles.
- **Univariate & Bivariate Exploration:** Analyze univariate distributions and investigate relationships between cocoa content and ratings.
- **Geographic & Manufacturer Profiling:** Evaluate chocolate quality across bean origin countries, company locations, and bean varieties.
- **Feature Engineering:** Categorize continuous cocoa percentages into functional business tiers.

- **Visualization Portfolio:** Produce high-impact, publication-ready visualizations in `ggplot2`.
- **Business Insights & Strategy:** Translate analytical findings into actionable recommendations for sourcing and production.

5. BUSINESS QUESTIONS

1. Which cocoa percentages receive the highest expert ratings?
2. Which countries produce the highest-rated cocoa beans and chocolate bars?
3. Which manufacturing companies consistently produce high-quality chocolate?
4. Does bean origin significantly influence overall chocolate quality?
5. What quality trends and rating trajectories exist across different review years?
6. Which specific cocoa percentage range yields the optimal balance of flavor and expert acclaim?

6. DATASET DESCRIPTION

The project analyzes the **Flavors of Cacao** dataset, comprising expert ratings and detailed profiles of chocolate bars produced globally.

Variable	Description & Characterization
`Company`	Name of the chocolate manufacturer
`Company Location`	Country where the chocolate manufacturer operates
`Specific Origin`	Specific designation, farm name, or bar origin name
`Broad Origin`	Broad geographic country of cocoa bean origin
`Cocoa Percent`	Percentage of cocoa content (e.g., 70%)
`Bean Type`	Variety of cocoa bean (e.g., Criollo, Forastero, Trinitario)
`Review Year`	Year the review was recorded
`Rating`	Expert numerical quality rating (scale 1 to 5)

7. SOFTWARE AND TECHNOLOGIES

Tool / Package	Analytical Purpose & Functionality
`R Environment`	Core statistical computing language and execution platform.
`RStudio`	Integrated Development Environment (IDE) for project workflows.

Tool / Package	Analytical Purpose & Functionality
`tidyverse`	Comprehensive ecosystem for data science (includes `dplyr`, `ggplot2`, etc.).
`janitor`	Automated cleaning of column names and frequency tables.
`dplyr`	Data manipulation, filtering, aggregation, and transformation.
`ggplot2`	Publication-grade data visualizations and thematic plots.
`skimr`	Comprehensive summary statistics and data profiling summaries.
`corrplot / GGally`	Multivariate correlation matrices and paired visual exploration.
`caret / randomForest`	Machine learning data preprocessing and predictive model preparation.
`DataExplorer / naniar`	Automated exploratory data profiling and missing value visualization.

8. PROJECT METHODOLOGY

Phase 1: Data Import & Workspace Setup

Load essential libraries (tidyverse, janitor, skimr), import raw CSV data, and inspect initial data structures and column types.

Deliverable: Imported dataset & structural summary report.

Phase 2: Data Cleaning and Standardization

Standardize column names via `janitor::clean_names()`, strip percentage symbols from `cocoa_percent`, convert variables to numeric/factor types, resolve missing values, and remove duplicates.

Deliverable: Cleaned dataset & Data Quality Assessment Report.

Phase 3: Exploratory Data Analysis (EDA)

Generate summary statistics using `skimr`, analyze univariate distributions of ratings and cocoa percentage, identify outliers, and evaluate missing value profiles with `naniar`.

Deliverable: EDA statistical summaries & initial diagnostic findings.

Phase 4: Feature Engineering

Convert string percentages into clean continuous floats and construct categorical bins (e.g., *Low: <60%*, *Medium: 60–70%*, *High: 70–80%*, *Very High: >80%*) for group comparisons.

Deliverable: Enhanced feature-engineered dataset.

Phase 5: Publication-Grade Data Visualization

Construct clear `ggplot2` visualizations including rating box plots across cocoa bins, origin maps/bar charts, company performance charts, yearly rating trends, and correlation heatmaps.

Deliverable: Complete visualization portfolio with technical interpretations.

Phase 6: Trend & Bivariate Analysis

Perform deep-dive comparative evaluations: Cocoa Percentage vs. Rating, Bean Origin vs. Rating, Company Location vs. Quality, and Review Year longitudinal dynamics.

Deliverable: Quantitative trend matrix & bivariate analysis chapter.

Phase 7: Business Insight Generation

Synthesize statistical output into executive insights, pinpointing optimal cocoa ranges (sweet spot), top-performing bean origins, and benchmark manufacturing practices.

Deliverable: Strategic Business Insight Summary.

Phase 8: Strategic Recommendations & Future Roadmap

Formulate actionable procurement guidelines, quality control benchmarks, bean variety selection rules, and roadmap predictive modeling using `randomForest`.

Deliverable: Strategic Action Plan & Future Technical Roadmap.

9. EXPECTED PROJECT DELIVERABLES

- ✔ Cleaned & Standardized Flavors of Cacao Dataset
- ✔ Data Quality & Completeness Assessment Report
- ✔ Comprehensive Exploratory Data Analysis (EDA) Report
- ✔ Univariate & Multivariate Statistical Summaries
- ✔ Feature-Engineered Dataset (with Cocoa Concentration Bins)
- ✔ Publication-Quality ggplot2 Data Visualizations
- ✔ Technical Data Interpretations
- ✔ Executive Business Interpretations & Insights
- ✔ Final Integrated Analytical Report
- ✔ Fully Reproducible R Analysis Scripts